

STELLA (YUXUAN) JIANG

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EDUCATION

B.S. in Computer Science, *GPA: 3.84*, University of California, Irvine Sep. 2023 – Expected Jun. 2027

SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS
Technologies: RESTful APIs, Microservices, Docker, Kubernetes, CI/CD, MySQL, NoSQL, Data Pipelines, Git, Linux
Awards/Certifications: AWS Certified Solutions Architect - Associate, IMMC Finalist

PROJECTS

- FlowCrusade: Event-Driven Workflow Orchestration System** Jan. 2026 – Present
- Architected a full-stack workflow orchestration system using **Python/Flask** and **React**, supporting concurrent multi-session execution via event-driven task scheduling.
 - Engineered a real-time Voice-to-Structure pipeline integrating streaming speech-to-text with LLM-based semantic parsing, producing standardized **JSON** task graphs for scalable workflow execution.
 - Designed a **DAG**-based dependency engine and secure **RESTful APIs** with modular JSON persistence, implementing failure handling and state reconciliation to ensure consistent execution and fault tolerance under concurrent workloads.
- AtlasSearch: Scalable Disk-Based Search Engine** Feb. 2026 – Mar. 2026
- Built a scalable search engine in **Python** implementing inverted index construction and ranked retrieval.
 - Designed a larger-than-memory indexing pipeline with partial index flushing and multi-stage index merging.
 - Implemented a ranking pipeline combining TF-IDF scoring with **PageRank**-based link analysis, achieving **sub-300ms** query latency across 20+ benchmark queries.
- Mosaic: Scalable Desktop Activity Processing Platform** Jan. 2026 – Feb. 2026
- Built a full-stack activity analytics system in **Python/Flask**, implementing scalable time-series pipelines to process **10K+** activity events per session with structured data processing.
 - Implemented a time-series segmentation engine with deterministic state transitions and similarity-based detection to improve stability under noisy activity streams.
 - Developed secure **RESTful APIs** integrating **Google Calendar** with **OAuth** authentication and built a **React** dashboard with real-time state synchronization.

WORK EXPERIENCE

- Software Engineer Intern, Beijing Jiuqi Software Co., Ltd** Aug. 2025 – Sep. 2025
- Migrated internal AI agents from Dify to Nüwa GPT, deploying containerized services using **Docker** and **Kubernetes** on **Alibaba Cloud**, reducing deployment time by 40% through **CI/CD** pipelines.
 - Developed **RESTful APIs** and backend services using **Python/Flask**, leveraging **Alibaba Cloud ECS** and **OSS** for scalable application deployment and data storage management.
 - Built internal automation tools and monitoring scripts using **Alibaba Cloud Function Compute**, improving team productivity and system reliability across distributed enterprise environments.
- Software Engineer Intern, Guolian Securities Co., Ltd** Aug. 2024 – Sep. 2024
- Designed **ETL pipelines** using **Python/SQL** on **Alibaba Cloud MaxCompute**, improving data accuracy by 25% for large-scale financial datasets through distributed computing.
 - Developed automated data validation scripts and **unit tests**, leveraging **Alibaba Cloud RDS** for database management and ensuring data integrity across multiple systems.
 - Built interactive analytics dashboards using **Pandas** and **Matplotlib** for financial data analysis.